Machine Learning Fundamentals

Week 1 – Exploratory Data Analysis (EDA)

Dataset: Titanic - Machine Learning from Disaster

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Track: Neurofive Machine Learning Track

Objective

The objective of this notebook is to explore the Titanic dataset, understand its structure, identify missing values, distinguish numerical and categorical features, and summarize the dataset before applying any machine learning models.

```python
import pandas as pd
import numpy as np

df = pd.read_csv(r"D:\Machine Learning Fundamentals Course\Week 1\
titanic\train.csv")

df.head()
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age
SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0
1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0
1
2                             Heikkinen, Miss. Laina  female  26.0
0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0
1
4                           Allen, Mr. William Henry    male  35.0
0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
```

```
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

```
df.describe()
```

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

```
print(df.shape)
```

```
(891, 12)
```

```
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
df.select_dtypes(include=np.number).columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp', 'Parch',
'Fare'], dtype='str')
```

```
df.select_dtypes(include=['object', 'str']).columns
```

```
Index(['Name', 'Sex', 'Ticket', 'Cabin', 'Embarked'], dtype='str')
```

Data Story

The Titanic dataset contains information about 891 passengers and 12 different features. The dataset includes both numerical and categorical variables such as age, fare, gender, passenger class, and embarkation point. Some columns, including Age, Cabin, and Embarked, contain missing values that will require preprocessing before training machine learning models. Overall, the dataset appears well-structured and provides sufficient information for predicting passenger survival.